

Physical Chemistry (II)

Lecture 14

Identical Particles
Variational Method



Fine Structure

1. Relativistic effect
2. Spin-orbit coupling

$$\hat{H}_0 \rightarrow \hat{H}_0 + \hat{H}_{\text{rel}} + \hat{H}_{\text{so}}$$

We can apply the perturbation theory
to obtain first-order correction to energy

Fine Structure of Hydrogen

The relativistic correction and spin-orbit coupling are of the same order

$$E_{\text{rel}} = -\frac{(E_n)^2}{2mc^2} \left(\frac{4n}{l + 1/2} - 3 \right)$$

$$E_{\text{so}} = \frac{(E_n)^2}{mc^2} \left\{ \frac{n[j(j+1) - l(l+1) - 3/4]}{l(l+1/2)(l+1)} \right\}$$

$$E_{\text{fs}} = E_{\text{rel}} + E_{\text{so}} = \frac{(E_n)^2}{2mc^2} \left(3 - \frac{4n}{j + 1/2} \right)$$

Atomic Term Symbol

$$^{2s+1}[l]_j$$

- Symbols for $[l]$

l	0	1	2	3	4	...
$[l]$	S	P	D	F	G	

- Selection rules for a hydrogen atom

$$\Delta l = \pm 1$$

$$\Delta j = 0, \pm 1$$

Precise Weak-Field Splitting

The energy splitting due to a weak magnetic field is

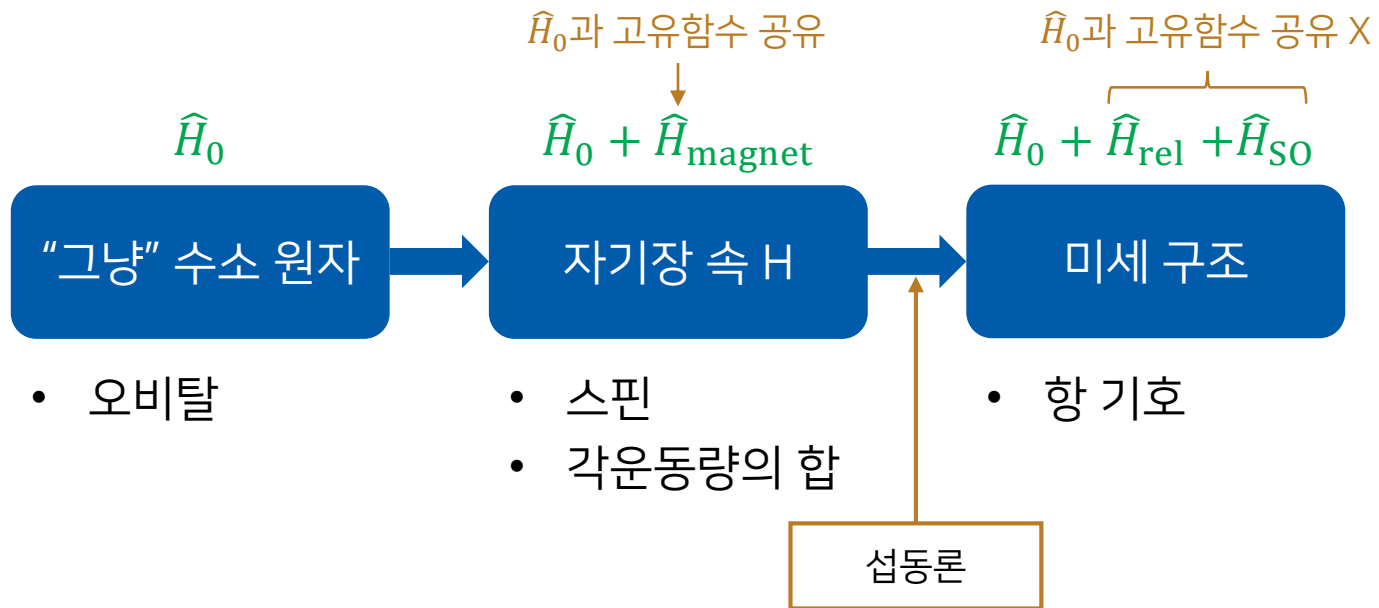
$$E_{\text{magnet}} = g(j, l)\beta_B m_j B$$

where $g(j, l)$ is the Landé g factor.

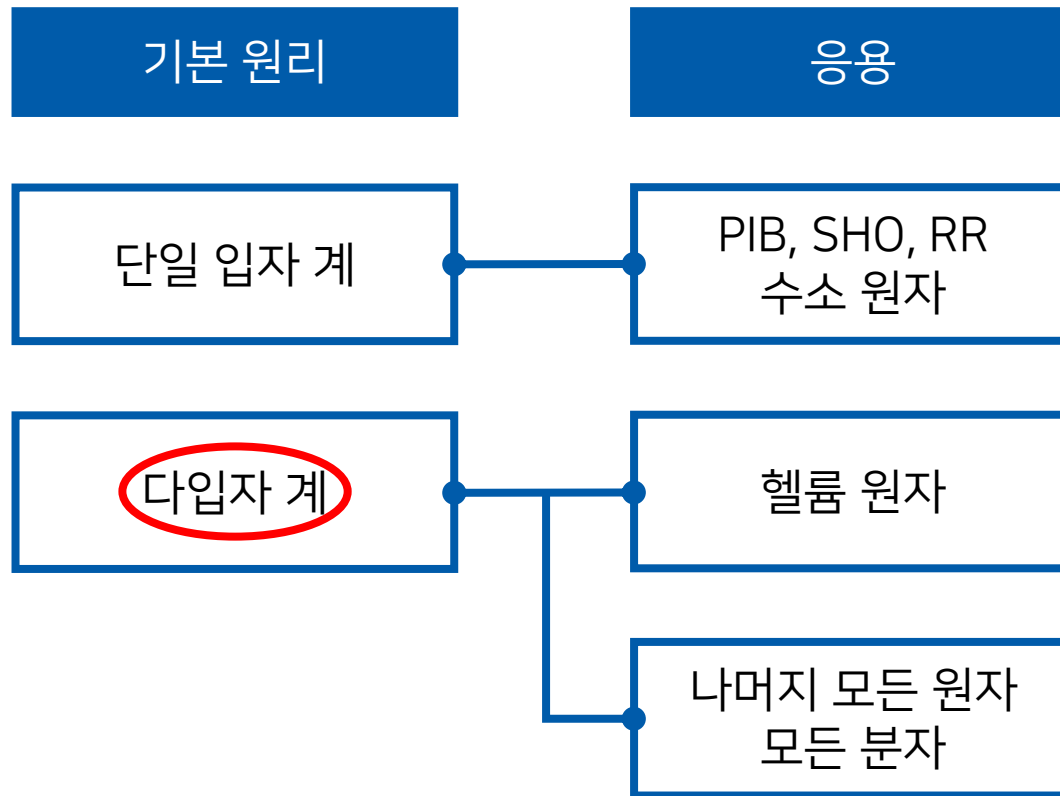
Considering the spin-orbit coupling, the energy levels are

$$E = -Z \frac{R_H}{n^2} + \frac{(E_n)^2}{2mc^2} \left(3 - \frac{4n}{j + 1/2} \right) + g(j, l)\beta_B m_j B$$

Where Are We: Hydrogen Atom

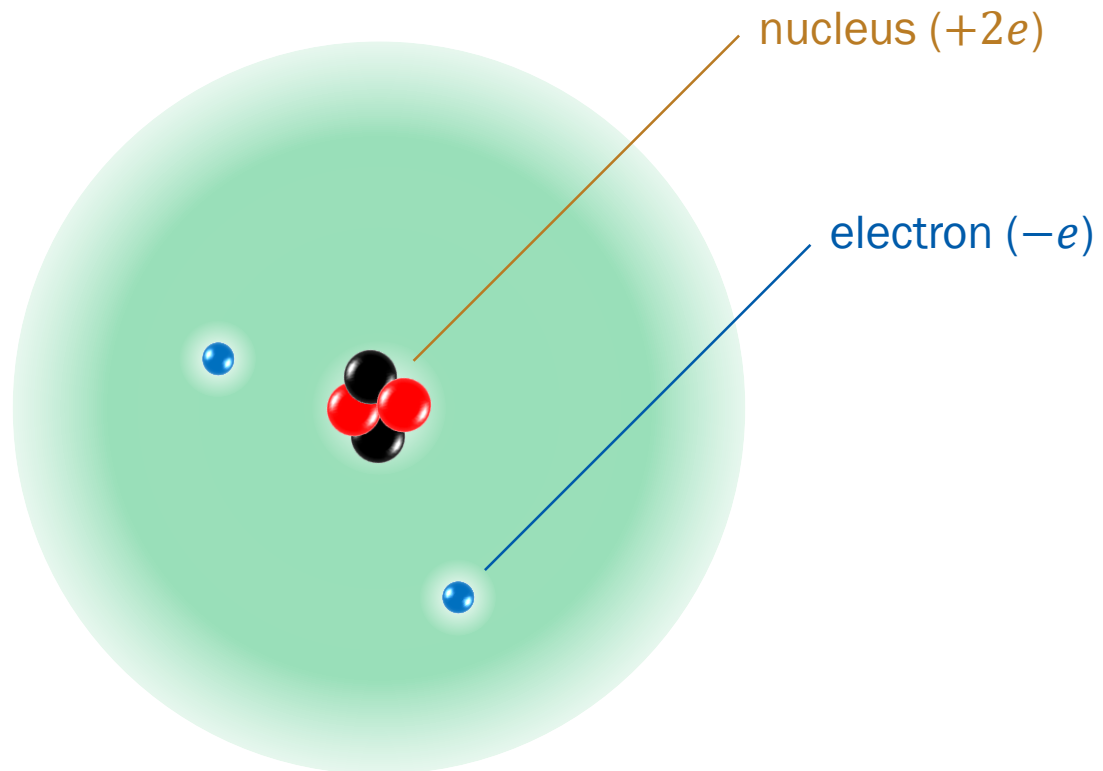


Where Are We?



RS 7.1
M 7.9

Helium Atom



Issues with Helium

- **Three-body problem:** not exactly solvable
 - Need approximation methods (e.g. perturbation theory)
- **Two electrons:** two identical particles
 - Meaning of wavefunctions
 - Indistinguishability: non-classical behaviors

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(*Griffith 5.1)

Two-Particle Wavefunction

- $\Psi(\vec{r}_1, \vec{r}_2, t)$: a complex function of ...
 - particle 1 position $\vec{r}_1$
 - particle 2 position $\vec{r}_2$
 - time t
- $|\Psi(\vec{r}_1, \vec{r}_2, t)|^2$: probability of finding particle 1 at point $\vec{r}_1$ and particle 2 at point $\vec{r}_2$ at time t

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Schrödinger Equation

$$i\hbar \frac{\partial \Psi(\vec{r}_1, \vec{r}_2, t)}{\partial t} = \hat{H} \Psi(\vec{r}_1, \vec{r}_2, t)$$

where

$$\hat{H} = -\frac{\hbar^2}{2m_1} \nabla_1^2 - \frac{\hbar^2}{2m_2} \nabla_2^2 + V(\vec{r}_1, \vec{r}_2, t)$$

If $V(\vec{r}_1, \vec{r}_2, t) = V(\vec{r}_1, \vec{r}_2)$, we obtain

$$\Psi(\vec{r}_1, \vec{r}_2, t) = \psi(\vec{r}_1, \vec{r}_2) e^{-iEt/\hbar}$$

$$\hat{H} \psi(\vec{r}_1, \vec{r}_2) = E \psi(\vec{r}_1, \vec{r}_2)$$

Identical Particles

- Indistinguishability: one cannot tell where a specific particle is

$$|\psi(\vec{r}_a, \vec{r}_b)|^2 = |\psi(\vec{r}_b, \vec{r}_a)|^2$$

- Fermions vs. bosons
 - Fermions: half-integer spin (electron, proton, neutron, ...)

$$\psi(\vec{r}_a, \vec{r}_b) = -\psi(\vec{r}_b, \vec{r}_a) \text{ (antisymmetric)}$$

- Bosons: integer spin (photon, helium nucleus, ...)

$$\psi(\vec{r}_a, \vec{r}_b) = \psi(\vec{r}_b, \vec{r}_a) \text{ (symmetric)}$$

Many-Particle Systems

For N identical particles,

- **Wavefunctions:** $\Psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N, t)$ and $\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)$
- **Indistinguishability:**

$$|\psi(\dots, \underset{\substack{\uparrow \\ \text{particle } i}}{\vec{r}_a}, \dots, \underset{\substack{\uparrow \\ \text{particle } j}}{\vec{r}_b}, \dots)|^2 = |\psi(\dots, \vec{r}_b, \dots, \vec{r}_a, \dots)|^2$$

- Fermions:

$$\psi(\dots, \vec{r}_a, \dots, \vec{r}_b, \dots) = -\psi(\dots, \vec{r}_b, \dots, \vec{r}_a, \dots)$$

- Bosons:

$$\psi(\dots, \vec{r}_a, \dots, \vec{r}_b, \dots) = \psi(\dots, \vec{r}_b, \dots, \vec{r}_a, \dots)$$

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Pauli Exclusion Principle

Two fermions cannot be detected at the same position

$$\psi(\cdots, \vec{r}_a, \cdots, \vec{r}_b, \cdots) = -\psi(\cdots, \vec{r}_b, \cdots, \vec{r}_a, \cdots)$$

$$\psi(\cdots, \vec{r}_a, \cdots, \vec{r}_a, \cdots) = -\psi(\cdots, \vec{r}_a, \cdots, \vec{r}_a, \cdots)$$

$$2\psi(\cdots, \vec{r}_a, \cdots, \vec{r}_a, \cdots) = 0$$

Hence,

$$|\psi(\cdots, \vec{r}_a, \cdots, \vec{r}_a, \cdots)|^2 = 0$$

Spin Wavefunction

- For two spins, we can write its wavefunction as

$$|m_{s,1}\rangle_1 |m_{s,2}\rangle_2$$

- Indistinguishability issue
 - $|+1/2\rangle_1 |+1/2\rangle_2$: indistinguishable, symmetric
 - $|-1/2\rangle_1 |-1/2\rangle_2$: indistinguishable, symmetric
 - $|+1/2\rangle_1 |-1/2\rangle_2$ or $|-1/2\rangle_1 |+1/2\rangle_2$: distinguishable!

How can we make “indistinguishable” spin wavefunctions?

Addition of Spin Angular Momenta

- Consider two spins: $|\pm 1/2\rangle_1$ and $|\pm 1/2\rangle_2$
- Define the total spin angular momentum $\vec{S}_{\text{tot}} = \vec{S}_1 + \vec{S}_2$:

$$\hat{S}_{\text{tot}}^2 |s_{\text{tot}} m_{s,\text{tot}}\rangle = s_{\text{tot}}(s_{\text{tot}} + 1)\hbar^2 |s_{\text{tot}} m_{s,\text{tot}}\rangle$$

$$\hat{S}_{z,\text{tot}} |s_{\text{tot}} m_{s,\text{tot}}\rangle = m_{s,\text{tot}}\hbar |s_{\text{tot}} m_{s,\text{tot}}\rangle$$

- Total spin quantum numbers

$$s_{\text{tot}} = (s_1 + s_2), (s_1 + s_2 - 1), \dots, |s_1 - s_2|$$

$$m_{s,\text{tot}} = -s_{\text{tot}}, -(s_{\text{tot}} - 1), \dots, s_{\text{tot}}$$

Two Electrons

$$(s_1, m_{s,1}) = \left(\frac{1}{2}, \pm\frac{1}{2}\right) \text{ and } (s_2, m_{s,2}) = \left(\frac{1}{2}, \pm\frac{1}{2}\right)$$

$$s_{\text{tot}} = (s_1 + s_2), (s_1 + s_2 - 1), \dots, |s_1 - s_2| = 1, 0$$

- $s_{\text{tot}} = 1: m_{s,\text{tot}} = -1, 0, 1$ (triplet)
- $s_{\text{tot}} = 0: m_{s,\text{tot}} = 0$ (singlet)

Spin Wavefunctions

- Triplet: symmetric

- $|s_{\text{tot}} = 1, m_{s,\text{tot}} = 1\rangle_{\text{tot}} = |+\frac{1}{2}\rangle_1 |+\frac{1}{2}\rangle_2$

- $|1, -1\rangle_{\text{tot}} = |-\frac{1}{2}\rangle_1 |-\frac{1}{2}\rangle_2$

- $|1, 0\rangle_{\text{tot}} = \frac{1}{\sqrt{2}} (|+\frac{1}{2}\rangle_1 |-\frac{1}{2}\rangle_2 + |-\frac{1}{2}\rangle_1 |+\frac{1}{2}\rangle_2)$

- Singlet: antisymmetric

- $|0, 0\rangle_{\text{tot}} = \frac{1}{\sqrt{2}} (|+\frac{1}{2}\rangle_1 |-\frac{1}{2}\rangle_2 - |-\frac{1}{2}\rangle_1 |+\frac{1}{2}\rangle_2)$

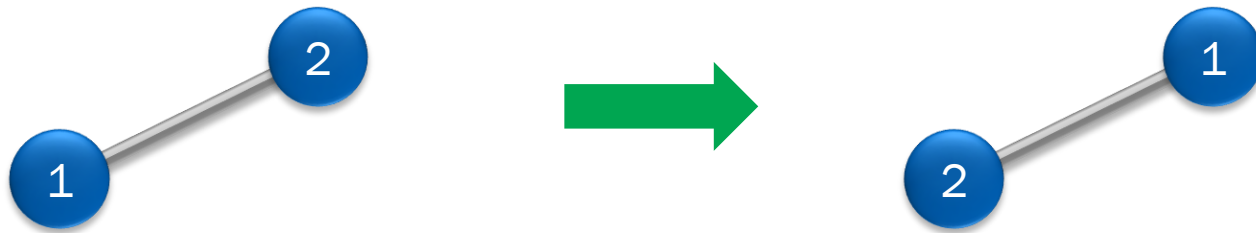
Total Two-Electron Wavefunctions

$$\psi(\vec{r}_1, \vec{r}_2) |s_{\text{tot}} m_{s,\text{tot}}\rangle_{\text{tot}}$$

- The total wavefunction must be antisymmetric
- If the spin wavefunction is triplet (symmetric),
the spatial wavefunction ψ must be antisymmetric
- If the spin wavefunction is singlet (antisymmetric),
the spatial wavefunction ψ must be symmetric

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Simple Model: Rigid Rotor



$$(x, y, z) \rightarrow (-x, -y, -z)$$

$$(r, \theta, \phi) \rightarrow (r, \pi - \theta, \phi + \pi)$$

Wavefunctions

- Spherical harmonics $Y_J^m(\theta, \phi)$
- Exchange of two atoms: $(\theta, \phi) \rightarrow (\pi - \theta, \phi + \pi)$

$$Y_0^0 = \frac{1}{2} \left(\frac{1}{\pi} \right)^{1/2} \rightarrow Y_0^0(\pi - \theta, \phi + \pi) = Y_0^0(\theta, \phi)$$

$$Y_1^0 = \frac{1}{2} \left(\frac{3}{\pi} \right)^{1/2} \cos \theta \rightarrow Y_1^0(\pi - \theta, \phi + \pi) = -Y_1^0(\theta, \phi)$$

$$Y_1^{\pm 1} = \mp \frac{1}{2} \left(\frac{3}{2\pi} \right)^{1/2} \sin \theta e^{\pm i\phi} \rightarrow Y_1^{\pm 1}(\pi - \theta, \phi + \pi) = -Y_1^{\pm 1}(\theta, \phi)$$

$$Y_2^0 = \frac{1}{4} \left(\frac{5}{\pi} \right)^{1/2} (3 \cos^2 \theta - 1) \rightarrow Y_2^0(\pi - \theta, \phi + \pi) = Y_2^0(\theta, \phi)$$

In General

- Odd J : antisymmetric

$$Y_J^m(\pi - \theta, \phi + \pi) = -Y_J^m(\theta, \phi)$$

- Even J : symmetric

$$Y_J^m(\pi - \theta, \phi + \pi) = Y_J^m(\theta, \phi)$$

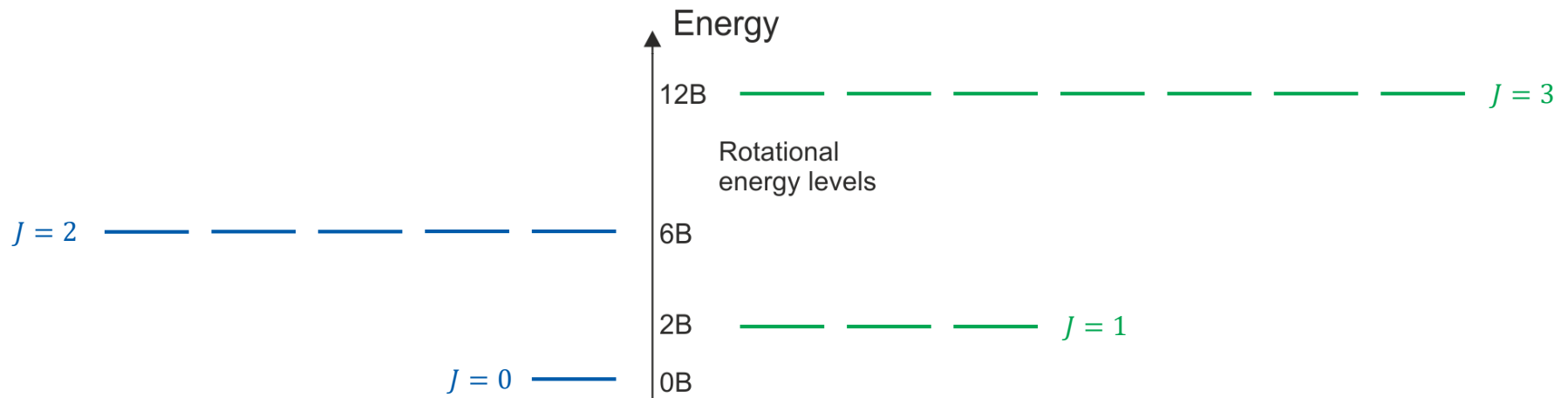
H₂ Molecule

- Consider only rotation and spin ($s = 1/2$ for a proton)
- Total wavefunction: $Y_J^m(\theta, \phi) |s_{\text{tot}} m_{s,\text{tot}}\rangle_{\text{tot}}$
 - Must be **antisymmetric**
- **Triplet** $|1 m_{s,\text{tot}}\rangle_{\text{tot}}$: symmetric (“ortho-hydrogen”)
 - Y_J^m must be antisymmetric: **odd J**
- **Singlet** $|0 0\rangle_{\text{tot}}$: antisymmetric (“para-hydrogen”)
 - Y_J^m must be symmetric: **even J**

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Energy Levels

(Number of states) $\times$ 3
due to spin states



para-hydrogen
(singlet, $s_{\text{tot}} = 0$)



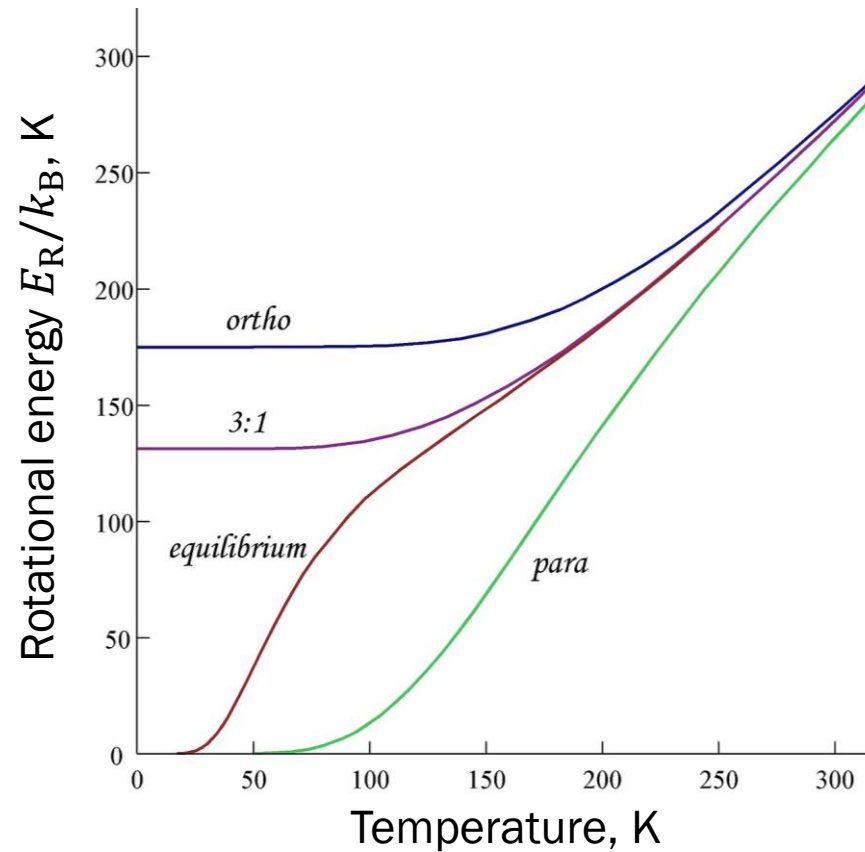
ortho-hydrogen
(triplet, $s_{\text{tot}} = 1$)

rare transition
(selection rule)



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Total Energy



(Image: https://commons.wikimedia.org/wiki/File:Ortho-para_H2_energies.jpg)

Approximation Methods

$\hat{H}\psi = E\psi$: No more exactly solvable

→ need approximation!

1. Perturbation theory
2. Variational Method

Variational Principle

Ground-state wavefunction ψ_0 and its energy E_0 satisfy

$$\hat{H}\psi_0 = E_0\psi_0$$

Multiplication by $\psi_0^*/\int \psi_0^*\psi_0 d\tau$ and integration lead to

$$E_0 = \frac{\int \psi_0^* \hat{H} \psi_0 d\tau}{\int \psi_0^* \psi_0 d\tau} = \frac{\langle \psi_0 | \hat{H} | \psi_0 \rangle}{\langle \psi_0 | \psi_0 \rangle}$$

For any function ϕ (“trial function”), the following holds:

$$E_\phi = \frac{\int \phi^* \hat{H} \phi d\tau}{\int \phi^* \phi d\tau} = \frac{\langle \phi | \hat{H} | \phi \rangle}{\langle \phi | \phi \rangle} \geq E_0$$

Proof

Since the set of wavefunctions $\{\psi_i\}$ is complete,

$$\phi = \sum_i c_i \psi_i = c_1 \psi_1 + c_2 \psi_2 + \dots$$

$$\int \phi^* \phi \, d\tau = \int (c_1^* \psi_1^* + c_2^* \psi_2^* + \dots)(c_1 \psi_1 + c_2 \psi_2 + \dots) \, d\tau$$

$$= \int (c_1^* c_1 \psi_1^* \psi_1 + c_2^* c_2 \psi_2^* \psi_2 + \dots + c_1^* c_2 \psi_1^* \psi_2 + \dots) \, d\tau$$

$$= c_1^* c_1 \int \cancel{\psi_1^* \psi_1} \, d\tau + c_2^* c_2 \int \cancel{\psi_2^* \psi_2} \, d\tau + \dots + c_1^* c_2 \int \cancel{\psi_1^* \psi_2} \, d\tau + \dots$$

$\underset{1}{}$
 $\underset{1}{}$
 $\underset{0}{}$

Proof

$$\phi = \sum_i c_i \psi_i = c_1 \psi_1 + c_2 \psi_2 + \dots$$

$$\int \phi^* \phi \, d\tau = c_1^* c_1 + c_2^* c_2 + \dots = \sum_i |c_i|^2$$

$$\int \phi^* \hat{H} \phi \, d\tau = \int (c_1^* \psi_1^* + c_2^* \psi_2^* + \dots) \hat{H} (c_1 \psi_1 + c_2 \psi_2 + \dots) \, d\tau$$

$$= \int (c_1^* \psi_1^* + c_2^* \psi_2^* + \dots) (c_1 E_1 \psi_1 + c_2 E_2 \psi_2 + \dots) \, d\tau$$

$$= c_1^* c_1 E_1 + c_2^* c_2 E_2 + \dots = \sum_i |c_i|^2 E_i$$

Proof

$$\phi = \sum_i c_i \psi_i = c_1 \psi_1 + c_2 \psi_2 + \dots$$

$$\int \phi^* \phi \, d\tau = \sum_i |c_i|^2, \quad \int \phi^* \hat{H} \phi \, d\tau = \sum_i |c_i|^2 E_i$$

Hence,

$$E_\phi = \frac{\int \phi^* \hat{H} \phi \, d\tau}{\int \phi^* \phi \, d\tau} = \frac{\sum_i |c_i|^2 E_i}{\sum_i |c_i|^2} \geq \frac{\sum_i |c_i|^2 E_0}{\sum_i |c_i|^2} = E_0$$

Trial Function

$$E_{\phi} \geq E_0$$

- May contain parameters (variational parameters)

$$\phi = f(\alpha, \beta, \gamma, \dots)$$

Minimize $E_{\phi}(\alpha, \beta, \gamma, \dots)$!

- May be a linear combination of basis functions

$$\phi = c_1 f_1 + c_2 f_2 + \dots + c_N f_N$$

Minimize $E_{\phi}(c_1, c_2, \dots, c_N)$!

Example: Hydrogen Atom

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

- Trial function: $\phi(\alpha) = e^{-\alpha r}$ (α : variational parameter)

$$\int \phi^* \hat{H} \phi \, d\tau = \frac{3\hbar^2 \pi^{3/2}}{4\sqrt{2}\mu\alpha^{1/2}} - \frac{e^2}{4\epsilon_0 \alpha}$$

$$\int \phi^* \phi \, d\tau = \left(\frac{\pi}{2\alpha}\right)^{3/2}$$

Example: Hydrogen Atom

$$E_{\phi}(\alpha) = \frac{\int \phi^* \hat{H} \phi \, d\tau}{\int \phi^* \phi \, d\tau} = \frac{3\hbar^2 \alpha}{2\mu} - \frac{e^2}{\epsilon_0} \left(\frac{\alpha}{2\pi^3} \right)^{1/2}$$

Minimizing $E_{\phi}(\alpha)$ by differentiation, we obtain

$$E_{\phi,\min} = -\frac{4}{3\pi} \left(\frac{\mu e^4}{16\pi^2 \epsilon_0^2 \hbar^2} \right) = -0.424 \left(\frac{\mu e^4}{16\pi^2 \epsilon_0^2 \hbar^2} \right)$$

- True value:

$$E_0 = -\frac{1}{2} \left(\frac{\mu e^4}{16\pi^2 \epsilon_0^2 \hbar^2} \right) = -0.500 \left(\frac{\mu e^4}{16\pi^2 \epsilon_0^2 \hbar^2} \right)$$

H-Atom Ground-State Energy

$$\phi(\{c_j, \alpha_j\}) = \sum_{j=1}^N c_j e^{-\alpha_j r^2}$$

N	$E_{\phi,\min}/(m_e e^4 / 16\pi^2 \varepsilon_0^2 \hbar^2)$
1	-0.42441
2	-0.48581
3	-0.49697
4	-0.49928
5	-0.49976
6	-0.49988
8	-0.49992
16	-0.49998

Number of Parameters

